BALIKTELAPH / TECHNICAL HANDOFF

BalikTelaPH Web Dashboard Data Flow Report

Information lineage, calculation definitions and mobile integration contract

18 October 2026
Reviewed source 5e21c45

For the web dashboard team, mobile developer, project administrators and research stakeholders.

Scope: implemented web behavior, clearly separated from synthetic data, placeholders and proposed integration requirements.

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1 Executive Summary

BalikTelaPH has two connected responsibilities: the mobile application records participation and textile exchange activity; the web dashboard manages, monitors, analyzes and governs the resulting records. Matching means connecting the same canonical records by IDs and agreed field meanings, not copying mobile screen totals into web cards.

The web implementation reads Supabase PostgreSQL through its Express API and converts database snake_case columns into TypeScript camelCase records. It then applies account permissions, selected cohorts and calculations before returning charts and tables. A shared database is necessary but not sufficient: mobile must create linked batches, measurements, transfers and recovery evidence in the structure the web actually reads. [S01-S07]

There are three separate information classes: operational records in Supabase, synthetic operational records deliberately inserted into those same tables, and a separate synthetic predictive dataset generated in the server and exported as bundled artifacts. A live database connection does not make a synthetic record real. Environmental equivalents are disabled, and several values still have explicit or implicit demonstration behavior.

Question	Current answer
Where do headline mass metrics come from?	material_batches, transfers, measurements and recovery_records; calculated by server/dashboardAnalytics.ts, not independently typed into each tile.
Where do mobile records match?	users.id, material_batches.id, listings.id, match_requests.id, transfers.id and facilities.id, with their referencing columns preserved across clients.
Does every mobile table already feed the dashboard?	No. Collection receipts, events, support records and several research tables exist in the contract but are not consumed by current analytics. See Sections 10 and 12.
Is predictive analytics using live mobile history?	No. It uses 9,600 generated synthetic observations on a fixed 120-week snapshot, separate from Supabase operational analytics.
Is mobile-to-web updating instantaneous?	No. There is server/client caching and no operational-data real-time subscription. A stale mounted page needs a data-fetch trigger.
What is the next integration gate?	Agree on identifiers, statuses, stage evidence, geographic vocabulary and denominators; then reconcile one controlled lifecycle end to end.

Review basis: local web source commit 5e21c45, reviewed 18 October 2026. This is an implementation data-lineage report, not a claim that production database contents, deployed functions or the separately maintained mobile application were inspected. No shared data was changed while producing it.

2 End to End Data Flow

Step	Data movement and responsibility
1 Participant action	A consumer, producer, recycler, NGO or collection operator records a listing, proposal, handover, receipt or recovery operation in the mobile app. An authorized web administrator can also register or govern records.

Step	Data movement and responsibility
2 Authorized write	Use the shared governed API, or a separately secured mobile backend/RPC with the same canonical contract and invariants. The mobile implementation route must be confirmed by its developer; the web repo does not establish it.
3 Shared persistence	Persist canonical IDs, kilogram weights, evidence stages and timestamps in Supabase public tables. Supabase Auth stores credentials; public.users stores the linked application profile and assigned scope.
4 Web source read	SupabaseStore calls get_dashboard_source_snapshot for analytics. If that optimization RPC is absent, it falls back to canonical table reads. Other registry endpoints can read tables directly through the same store. [S03, S06]
5 Translation and joins	Codecs map snake_case to camelCase. Transfers are enriched with flow_events and latest stage-specific measurements; listings can inherit missing material metadata from their linked batch. [S04]
6 Authorization and selection	The Express API reloads current session/account permissions and selects permitted batches, then applies reporting dates and global filters. Aggregate-only responses suppress individual transaction/stakeholder lists. [S07, S08]
7 Derived data	Dashboard builder calculates masses, counts, rates, facility allocations, regional groups and engagement summaries. Report generation reuses the same cohort and aggregation logic. [S02, S12]
8 Presentation	React consumes /api/analytics/dashboard or the relevant registry endpoint. Chart.js renders analytics; Leaflet renders reference geography. Canonical kg are converted to metric tons at the display boundary.

Main operational chain: users + organizations -> material_batches -> listings -> match_requests -> transfers -> measurements + flow_events -> recovery_records + reward_ledger -> scoped API calculations -> web charts, registries and reports. Each arrow is a relationship, not an extra quantity to add.

Independent predictive chain: deterministic synthetic generator -> 9,600 event observations -> 120 weekly aggregates -> chronological training/validation/test -> forecast and selection models -> /api/analytics/predictive -> Advanced Analytics. Supabase operational tables are not an input to this chain today.

3 Source Ownership and Record Grain

Canonical source	Who originates it	What one record means
auth.users + users	Supabase identity flow; participant profile; administrator approval	One Auth identity and its linked application profile. Role/scope are administrator-governed, not inferred from profile text.
organizations / facilities	Administrator and authorized institutional representatives	One institution or one physical/program facility; capacity is a declared monthly capability, not a measured throughput total.

Canonical source	Who originates it	What one record means
material_batches	Mobile participant or authorized administrative registration	One identified quantity of textile material. Count its declared mass once even if it appears in multiple listings or transfers.
listings	Mobile availability/exchange workflow	One marketplace publication linked by batch_id. Listing availability is not proof of collection, receipt or recycling.
match_requests	Prospective recipient; provider response	One proposal on one listing. The web API calls this an offer; it is not a separate offers table.
transfers	Accepted proposal / governed workflow	One agreement/custody transfer linked to a batch and listing, with provider, recipient and outcome.
measurements	Named recorder and, where applicable, verifier	One mass observation at one lifecycle stage and time. Declaration, handover, receipt and validation must stay separate.
flow_events	Governed lifecycle action	One sequenced status/event observation with actor, timestamp, location and optional measurement reference.
recovery_records	Facility/collection/recovery operator; authorized validator	One stage input/output balance for a batch at a facility. Stage totals are not automatically unique mass.
reward_ledger	Reward-issuing workflow after eligible completion	One non-cash points entry linked to a user and, where applicable, transfer. Not textile mass or revenue.
system_interactions	Instrumented client/server or summary-producing job	Current analytics expect event-count summary rows with properties.count, not arbitrary raw click rows.
activity_logs / configurations	Governed API actions / administrator	One audit entry or versioned configuration. They explain decisions and definitions; they do not represent extra textile mass.

Do not join organizations by display name, users by name/email, or materials by title. Names and emails can change or be duplicated. Use canonical IDs; preserve the source profile ID when an Auth trigger already created it. The persistence contract uses text business IDs; auth.users.id is a UUID linked by users.auth_user_id. [S05, S20]

3.1 Where the synthetic operational figures originate

The original January-September 2026 fixture is generated in src/data/mockData.ts, not collected from participants. Preset national monthly declared masses range from 19,200 kg in January to 49,300 kg in September. Each month is allocated across four region shares (Metro Manila 48%, Central Visayas 20%, Western Visayas 14%, Davao 18%) and six textile shares. Collection and sorting percentages increase by formula; category-specific reuse/recycling ratios create the outcome masses. These assumptions intentionally create a coherent demonstration trend, not measured regional performance. [S25]

buildOperationalDemo converts those fixture figures into linked SYN- batches, declaration/handover/receipt/validation measurements, listings, accepted proposals, completed transfers, Collection/Sorting recovery balances, flow events and reward entries. Engagement is generated separately from

preset monthly active-user totals and formula-based session/submission/action/export summaries. Thus engagement totals are not a count of the small synthetic user registry. The initial Supabase dashboard seed persists the resulting records; the web then calculates cards from those tables. It does not use every legacy mock export as a live frontend fallback. [S15]

October is a separate additive synthetic scenario covering 1-16 October 2026: 24 batches based on September fixture rows, with mass scaled by 1.08 and constrained to coherent collected/sorted/outcome balances. Engagement scales selected September summary counts by 1.03 or 1.08 and uses a 16 October timestamp. It reuses fictional SYN registry records. It is a month-to-date illustration, not a full-month projection or observed October growth. These seeding paths describe intended fixture provenance; they are not a fresh inventory of the deployed database. [S16]

4 Dashboard Information Lineage

The entries below describe the current implementation. Column names refer to Supabase; names after an arrow are the web/API aliases. Operational analytics share the selected batch-registration cohort unless a different time basis is stated. [S02, S04, S08]

4.1 Overview and Waste Analytics

Information	Source fields	Calculation and mobile requirement
Generated / declared textile waste	material_batches.declared_weight_kg; id; created_at; current_status	Sum declared mass once per selected non-archived batch. This means platform-declared textile material, not total Philippine waste or total manufacturer output. Mobile must create the batch, not only a listing.
Textile waste collected	recovery_records.stage = Collection; output_kg; batch_id; status	Sum Collection outputs for selected batches, excluding reversed records and future-created recovery rows. A listing, receipt measurement or collection_receipts row alone does not update this KPI. Recorded and flagged recovery rows are not otherwise excluded today.
Sorted material	recovery_records.stage = Sorting; output_kg	Sum non-reversed Sorting outputs. Mobile must write the stage balance explicitly; processing/receipt records are not inferred as sorting.
Textile waste recycled	transfers.status = completed; circular_outcome = Recycling; measurements.stage = validation; weight_kg	Sum latest validation weight for completed Recycling transfers. Status alone without linked validation measurement contributes no verified mass.
Reused / Bartered	Completed transfers with Reuse, Repair, Redistribution or Upcycling outcome; latest validation weight	Sum these outcome weights. The current label includes more than literal barter; transfer_type is not used by this classifier. Collection, Recovery and Diversion outcomes are not included in these reuse/recycling totals.
Diverted from Landfill tile	Verified reused mass + verified recycled mass	Operational proxy equal to reused + recycled. It is not a validated landfill-avoidance calculation; environmental landfill equivalents remain disabled. Rename/qualify this label before claiming verified environmental impact.
Diversion / circularity rate	Reused + recycled mass; declared batch mass	100 x (reused + recycled) / declared mass. Uses a different denominator from category recovery and sorting yield. Zero denominator should be interpreted as unavailable.
Prior-period change	Same calculation on prior selected registration cohort	Mass/count delta = 100 x (current - previous) / previous. Rate changes use percentage points, not relative percent change. No positive baseline is labeled as such, not infinity.
Total transactions KPI	transfers.status = completed; linked batch_id	Count completed scoped transfers, not all proposals, all transfers, wallet entries or lifecycle events. A missing verification weight can still count as a completed row and produce a quality finding.

Information	Source fields	Calculation and mobile requirement
Active producers	Selected batch source_user_id joined to users.role = Producer	Distinct source users per batch region, then sum regional counts. This is participation in the cohort, not login activity; the same user operating in multiple regions can be counted more than once platform-wide.
Active recyclers	Selected transfer recipient_id joined to users.role = Recycler	Distinct recipients per batch region across matching transfers, including unfinished ones; regional counts are summed. Not a unique platform MAU or a count of all registered recyclers.
Waste trend and featured sparkline	Same declared/collection/reuse/recycling calculations grouped by batch created_at month	Months use Asia/Manila batch registration, not measurement/completion month. A later completion can change an older registration-month bar. Partial current months are not forecasts.
Textile category mix	material_batches.material_category; Collection outputs	Category mass is collected mass, not declared or recycled mass. Doughnut share = category collected mass / all collected mass in the selected scope.
Category recovery rate	Completed reused + recycled mass in category; category collected mass	100 x category recovered / category collected. This denominator differs from the overall diversion rate. Current helper returns 0 when collected mass is zero.
Circular textile flow	Declared batches, Collection/Sorting outputs, completed outcome weights	Generated, collected, sorted, reused and recycled are successive/branching stages; never add them as total waste. Residual = max(0, declared - reused - recycled). It includes uncollected, unverified and pending material, not just physical disposal/loss.
Recent transactions	Latest scoped transfers; users; linked batch location; flow_events	Sort by updated_at; show five rows. Display weight chooses validation, then receipt, then agreed mass. An unfinished row is not verified diversion.
Sample insights	regionalActivity.wasteGapKg; category recovery rates; facility utilization	Deterministic ranking: largest gap, lowest recovery and lowest-utilization facility. Spare capacity = 100 - utilization when <=100. These are simple cues, not a prescriptive or ML system.

4.2 System Oversight and Environmental Information

Information	Source	Interpretation
Batch count / declared material	Unique selected material_batches.id and declared_weight_kg	Same batch population as program analytics. A batch can have multiple measurements and transfers without increasing its declared mass count.
Verified completion	All completed transfers, latest validation measurement	Operational helper sums completed verified mass regardless of circular_outcome. This can differ from reused + recycled if mobile uses other outcomes. [S24]
Measured material	All selected measurements with weight_status = Measured	Sum of stage observations; the same material may be weighed more than once. It is not unique collected mass.

Information	Source	Interpretation
Active transfers	Transfers not Completed, Cancelled or Rejected	Includes Reserved, Handed over, Received, Validated and Disputed states after decoding.
Mass balance	Declared batches; handover/receipt/validation measurements; transfers.discrepancy_kg	Declared counted once; handover and receipt summed across transfers; verified completion uses completed only. Discrepancy is signed receipt minus handover, falling back to agreed.
Quality flags in oversight	Transfer discrepancy_kg and batch weight_status	Oversight discrepancy count uses absolute variance >0.01 kg. It is not the configured percentage-threshold count shown by Data Quality.
Circularity and sorting yield	Completed reused + recycled weights divided by declared or sorted mass	Circularity = recovered / declared; sorting yield = recovered / sorted. Percentages can exceed 100 when evidence is inconsistent; they are not silently capped.
CO2e, water and landfill-equivalent impact	No approved factors applied; API values null	Disabled pending approved methodology. Do not derive them from mobile My Impact totals or treat null as zero impact.

4.3 Listings Materials and Traceability

Information	Source and relationship	How mobile should match it
Listing title, description and owner	listings.title, description, owner_name, owner_id, owner_organization_id	Save owner_id as users.id. Names are cached display fields; keep canonical relationships even if names change. Listing oversight uses local search/status/region/material/pathway/evidence/source/demo filters, not the global cohort filter.
Listing material, condition and quantity	category -> textileType; condition -> quality; weight_kg -> quantityKg, falling back to available_weight_kg	Keep weight_kg and availability semantics aligned; existing weight_kg takes precedence. Missing material/location metadata can fall back to the linked batch. reserved_kg is not itself the displayed quantity.
Listing counts and availability	Decoded listing status: Open, Matched, Reserved, Completed, Closed	Counts come from locally filtered listings. These counts do not prove physical transfer or recovery. Persist canonical lowercase statuses; see Section 7.
Material identity and provenance	material_batches.id, source_user_id, source_organization_id, material_category, waste_source, waste_form, condition	Use the original batch ID throughout the chain. Keep a distinct source user and source organization; do not substitute a recycler or the current viewer for the original source.
Material current state and location	current_weight_kg, weight_status, current_custodian_id, current_status, intended_pathway/final_pathway, location JSON	Preserve declared_weight_kg when a receiving measurement changes current mass. final_pathway overrides intended_pathway in decoding. Batch region/city comes from location.region and location.city.

Information	Source and relationship	How mobile should match it
Registry verified mass	Current weight of batches whose weight_status = Verified	This is a stock/evidence-state sum, not the same as completed transfer outcome mass. Do not expect it to equal the recycled KPI automatically.
Material measurement history	measurements.batch_id -> material_batches.id	All stage rows appear under the batch. Include recorder, verifier, method, measured_at and note, with a separate ID per observation.
Offers tab	match_requests.requester_id -> recipientId; provider_id; listing_id; requested_kg; terms; incentive_points	The requester is the prospective recipient, not the provider. Accepted proposals must be linked to transfers through match_request_id; acceptance alone is not completion. The offers view loads its own offer list rather than applying every global analytics filter.
Transfer agreement	transfers.id, batch_id, listing_id, match_request_id, provider_id, recipient_id, agreed_weight_kg, agreed_terms	Every reference points to the same canonical row used by mobile. Whole agreement and physical measurements are separate concepts.
Handover, receipt and verified columns	Latest measurements per transfer_id and stage handover / receipt / validation	Write these measurement rows; there are no standalone handover_weight_kg/receipt_weight_kg/verified_weight_kg columns in the canonical transfers table. Sorting by measured_at selects the latest observation.
Lifecycle timeline	flow_events.transfer_id; sequence; stage; actor_id/name; occurred_at; event_status; measurement_id; location; note	Append structured events for each action; do not provide only a free-text transaction history. Measurement links explain which weight supports each event.
Discrepancy and reward status	transfers.discrepancy_kg, discrepancy_reason, reward_issued_at; reward_ledger	Receipt variance is signed; document a reason. reward_issued_at is an issuance marker; points must reconcile with ledger entries and must not be issued twice.

4.4 Stakeholders Facilities and Geography

Information	Source and rule	Important boundary
Stakeholder directory and groups	Participating source/provider/recipient users joined by users.id; role selects Producer, Recycler, Consumer or NGO tab	Only users involved in the filtered cohort appear, not all registered users. Upcycler and LGU roles exist but are not separate directory tabs today. Aggregate-only access suppresses individual profiles.
Stakeholder generated / recovered	Generated = own source batch declarations; recovered = completed verified transfers received by user	A provider may show zero recovered even when its textiles are recovered downstream. Do not add stakeholder source and recipient masses as new material.

Information	Source and rule	Important boundary
Stakeholder completion / activity	Completed linked transfers / all linked transfers; count all provider-or-recipient involvements	Overall profile activity can count a transfer under both participants; it is not a unique transaction count.
Sustainability and verification	users.profile.sustainability_interests; verification_status; created_at	These are profile claims, assigned verification states and account creation date, not independently verified impact calculations.
Trust / reliability	users.trust_placeholder, default 50 when absent	Stored demonstration score displayed as /100. No validated algorithm or reliability model; never use as an approval or risk decision automatically.
Stakeholder traceable material	Sum displayed weights of matching transaction history	History can include unfinished transfers and uses verified/receipt/agreed fallback; this is activity mass, not necessarily unique or completed recovered mass.
Facility identity / materials / status	facilities.name, city, region, categories, status, organization_id	Analytical facility table includes non-inactive facilities that have recovery rows for selected batches. It is not a directory count of every facility in Supabase. Status is stored, not recalculated from utilization.
Facility period capacity	monthly_capacity_kg x selected window capacityMonths	CapacityMonths sums actual calendar-month fractions. Changes to today's capacity apply to the current calculation; no historical capacity-version reconstruction is implemented.
Facility intake	Earliest non-reversed recovery row for each selected batch at that facility	If earliest stage is Collection use output_kg; otherwise use input_kg. This avoids adding every processing stage, but repeated independent visits of one batch to one facility are collapsed. current_intake_kg is not the analytical intake source.
Facility processed mass	Completed verified transfers for batches attributed to their latest recorded recovery facility	Attributed once per batch destination facility, not every facility visited. It includes all completed outcomes, not only Recycling, despite some UI wording.
Utilization / network utilization	Facility intake / prorated capacity; network = total intake / total capacity	Weighted by capacity, not average of facility percentages. Not historical calendar throughput utilization, because intake is selected by batch cohort. Values above 100 remain visible.
Processing time	Fixed 2.4 days in dashboard builder	Placeholder, not calculated from received/completed timestamps. Aggregate facility groups hide this value.

Information	Source and rule	Important boundary
Regional mass / recovery gap	Group selected batches by location.region, then use identical mass formulas	Recycling is assigned to batch origin region, not necessarily the recycler/facility region. Gap = max(0, declared - reused - recycled). Regional masses should reconcile to platform mass.
Collection points / recyclers on map	Collection points = active/non-inactive facility rows of type Collection point in batch region; recyclers = distinct transfer recipients of Recycler role per origin region	Collection-point counts are registry counts, not counts of physical collection events. Recycler involvements can recur across regions.
Map locations and boundaries	Bundled Natural Earth country geography; regional-anchors.json; Leaflet	Pins indicate reference regional locations, not live GPS, exact participant addresses, routes or regional boundary polygons. Unrecognized region names can appear in totals without a pin.
Regional share percentage	Selected regional metric / sum of same metric across filtered regions	Collection, Recycling and Recovery gap toggles change this denominator. A map selection only highlights/locates a region; the global Region filter controls the analytics cohort.

4.5 Engagement Barter and Diagnostic Views

Information	Current source and formula	Mobile matching rule
Monthly active users	Sum system_interactions.properties.count for event_name active_users in the final selected Manila calendar month	Summary rows must describe disjoint user populations if summed. Not COUNT(DISTINCT user_id), not users table size, and not sessions table count. Missing final-month summary is unavailable.
Sessions and average duration	Sum count for sessions; duration = sum(averageSessionMinutes x count) / sum(count)	Send count and weighted-average inputs in properties with region/city. Raw event rows without count contribute zero; no automatic session-to-engagement rollup exists.
Waste submissions / barter actions / exports	Sum properties.count for waste_submissions, barter_actions and report_exports	Operational actions and report.exported audit logs do not automatically populate these summaries. Mobile and web need an agreed instrumentation/aggregation writer.
Engagement filter scope	occurred_at reporting window, properties.region/city and actor scope	Material, pathway, weight, facility and other batch-specific filters are not applied to these summaries. Do not claim every global filter affects engagement.

Information	Current source and formula	Mobile matching rule
Barter metrics returned by API	Listings for selected batches; matches = all linked proposal rows; validated = transfer status Validated or Completed; completed = Completed transfers; rewards = linked Earned ledger points	These are not an exclusive funnel and are not filtered to transfer_type barter. Counts use batch-cohort relationships, not each listing/offer/reward creation date.
Barter Activity presentation	BarterActivity.tsx exists; barterMetrics returned by analytics	Component is not mounted in the current DashboardApp page tree. Treat this as available code/API data, not an active standalone dashboard page.
Diagnostic analytics	Same operational diversion rate, transaction completion, category recovery, facility utilization and charts	No additional model or independent dataset. Aggregate-only users have empty detailed transactions; some diagnostic zero/empty displays require careful interpretation.
Rules-based listing match score	API matchScore: base 20 + configured category/pathway/region/condition/quantity weights, capped at 100; candidates >=35	Optional mobile/API recommendation logic, not predictive category probability, not a trained model and not the barterMetrics.matches count.

4.6 Governance Quality Reports and Administration

Module / information	Actual origin	How it is maintained
Data quality findings	Computed from scoped batches/transfers and active configuration; not read from data_quality_issues	Estimated mass warning; Reserved >7 days warning; discrepancy threshold critical; Completed with null validation weight critical. Findings may overlap on the same material.
Validation coverage	Verified batch count / selected unique batch count	Count-based percentage, not verified kilograms / declared kilograms. Batch evidence state must match mobile measurements.
Discrepancy threshold	configurations.value.discrepancyThresholdPercent, default 5	Critical when abs(discrepancy)/handover mass >= threshold, agreed mass as fallback. Compared using unrounded values.
Governance and policy readiness	Current role/scope; configuration.methodologyVersion; fixed server policy text	Methodology/approval caveats are not dynamically read from methodologies, nor proof of statutory compliance.
Material Flow Summary report	Same cohort and mass aggregation as charts, grouped by region/material plus TOTAL	Exports include timeframe, filters, scope, generation timestamp, unit and definitions. CSV/print use full report rows, not current UI pagination.
Data Quality Summary report	Same computed quality rules as Data Quality page	Contains evidence counts, threshold and grouped findings; total findings do not mean unique affected batches.

Module / information	Actual origin	How it is maintained
Organization Participation report	organizations + users + source batches and linked transfers	Counts participating organizations; source declarations and completed verified receipts are distinct roles. Aggregate-only exports group by region/type and omit identifying names.
Environmental Methodology Status report	Configuration version + operational mass; environmental fields not calculated	It reports readiness and measured inputs, not invented CO2e/water savings.
Report audit / history	activity_logs action report.exported for successful export requests	report_exports table is not populated by this report route. Preview is not an export audit. system_interactions export summaries are a different source.
User account management	users + Supabase Auth + auth_account_security; users API	Admin invitations/approval/role/scope/status are authoritative. Mobile profile edits must not overwrite institutional role, scope, or verification.
Organization and facility administration	organizations and facilities through role-checked management API	Administrative edits save audit records. They change master data and can affect later analytics; they do not manufacture operational measurements.
My Profile and sessions	Current users record; profile JSON; sessions metadata	Profile fields are user-entered/assigned, not usage metrics. Supabase credentials and hashed refresh tokens are not exposed as dashboard information.
Settings and reference data	Active configurations row with key platform; value JSON	Taxonomies, discrepancy threshold, reward rate, matching weights and methodology version. UI reward rate per ton converts to stored credits per kg.
Audit trail	activity_logs actor_id, action, entity_type/id, details, occurred_at	API actions append audit evidence; aggregate-only responses redact details. Do not use logs as a second material ledger.
Public homepage / partner logos	Static publicContent.ts, local image assets, approved external links	Not a live data feed, partner database export or source of dashboard KPIs. No protected operational records are requested by the homepage.

5 Predictive Analytics Data Flow

Advanced Analytics > Predictive is a separate proof of concept. Runtime imports syntheticDataset from the deterministic generator in server/analytics/synthetic.ts; it does not read observations.csv on every request and does not query mobile operational tables. The CSV, manifest, evaluation and dictionary are reproducible exports of that generator. Editing the CSV alone will not change runtime forecasts. [S17-S19, S27]

Output / input	Origin and computation
Training data	Seed 20261016; 9,600 generated linked declaration/collection/offer-response observations; 80 records per week across four regions and six categories. No real participant identities.
Historical window	120 complete Monday-Sunday UTC weeks from 24 June 2024 through 11 October 2026. Four forecast weeks start 12, 19, 26 October and 2 November 2026. Fixed snapshot, not next four weeks relative to each page visit.
Partition	84 training weeks / 6,720 records; 18 validation weeks / 1,440 records; 18 test weeks / 1,440 records, chronologically 70/15/15. 9,600 records aggregate to 120 weekly time points.
Weekly collection forecast	Sum collectedKg per week; compare last value, trailing 8-week mean, trailing 26-week OLS trend and 52-week seasonal naive. Lowest validation MAE chooses the method. Predicted collection is constrained not to exceed declared mass.
Projected textile waste	Forecast weekly declaredKg using the same baseline comparison. Means projected platform-declared mass, not all textile waste in the Philippines or a validated national forecast.
Four-week totals	Sum four weekly forecasts separately for declared and collected mass. These are not additional independent masses; no total confidence interval is claimed.
Weekly uncertainty bands	Per-horizon 80th percentile absolute validation errors; nominal empirical ranges with actual test coverage reported. Not guaranteed 80% confidence.
MAE / WAPE / bias / coverage	MAE = sum absolute errors / predictions; WAPE = sum absolute errors / sum observed mass; bias = mean(predicted - observed); coverage = held-out observations inside displayed ranges / observations.
Recycler selection tendency	Category-conditioned Beta(1,1) smoothed probability = (selected offers + 1)/(offered observations + 2), using validation-selected 8/13/26/52-week recency. Not individual recycler prediction or a distribution that sums to 100%.
Actual selected share	Selected observations in category / all selected observations in chosen window. Different denominator from category selection probability. Selection is not proof of receipt or recycling.
Selection evaluation	Log loss and Brier score on held-out responses, compared with a category-blind pooled baseline. Lower error is better; synthetic success does not establish real-world accuracy.
Scope gates	At least 600 observations and 104 active weeks per selected region/material scope. Region permissions are enforced. Insufficient scopes return no prediction; these are PoC heuristics, not approved research sufficiency thresholds.
Filter and export behavior	Dedicated region/material/history controls, not global operational date filters. History selector changes displayed history only. Export requires separate permission and includes only permitted synthetic rows. Model runs cache in process; no model_runs writes.

Future live-model mapping is a separate task: declaration target from unique batch declarations at registration time; collection target from deduplicated Collection evidence at actual collection time; selection target from eligible resolved proposals, including non-selected outcomes and known decision times. Preserve observation timing and avoid using final outcome fields as predictors available before the decision. First choose whether the live target is a calendar-event series or a registration-cohort series; the current operational charts and synthetic weekly model use different time concepts.

6 Exact Mobile to Web Field Mapping

Database column names are the shared contract. The web aliases below are internal/API names, not new database columns. Mobile can keep its own UI names if its adapter consistently translates them. Do not create an offers table or write camelCase dashboard payloads directly into snake_case PostgreSQL tables. [S03-S05]

6.1 Identity and institution

Supabase column	Web alias	Meaning / matching rule
users.id / auth_user_id	id / authUserId	Business profile ID / Supabase Auth UUID bridge. Foreign keys use the correct namespace.
users.name / email / role	fullName / email / role	Identity display and authorization role. stakeholder alone does not grant access.
users.organization_id / organization	organizationId / organization	Canonical institutional ID / display label.
users.account_status / verification_status	accountStatus / verification	Lowercase persisted states decoded for UI. See normalization caveats.
users.scope.organization_ids, facility_ids, regions, can_export, aggregate_only	scope.organizationIds, facilityIds, regions, canExport, aggregateOnly	Administrator-controlled JSON access boundaries; preserve existing values in mobile updates.
users.profile.age_group, organization_size, production_volume_kg, recycling_practices, accepted_textiles, sustainability_interests	profile.ageGroup, organizationSize, productionVolumeKg, recyclingPractices, acceptedTextiles, sustainabilityInterests	Self-reported profile information; production_volume_kg is not generated-waste KPI input.
users.trust_placeholder / created_at	trustPlaceholder / joinedAt	Demonstration score / account creation timestamp.
organizations.id, name, type, record_label, verification_status, status, city, region	id, name, type, recordLabel, verification, status, city, region	Institution master record. service_areas -> serviceAreas; accepted_materials -> acceptedMaterials; contact_email -> contactEmail.

6.2 Materials and listings

Supabase column	Web alias	Meaning / matching rule
material_batches.id / source_user_id / source_organization_id	id / sourceUserId / sourceOrganizationId	Stable material identity and original provenance.
material_category / waste_source / waste_form / condition	materialCategory / wasteSource / wasteForm / condition	Exact taxonomy values shared by both apps.
intended_pathway / final_pathway	circularPathway	Decoder prefers final_pathway when non-null. Do not confuse intended Recycling with completed Recycling.
declared_weight_kg / current_weight_kg / weight_status	declaredWeightKg / currentWeightKg / weightStatus	Original declaration / latest batch state / Estimated, Measured or Verified.
location.city / location.region	city / region	Nested batch JSON, not top-level batch city/region columns.
current_status / current_custodian_id / archived_at	currentStatus / currentCustodianId / archivedAt	Custody and archival; analytics exclude Archived.
secondary_quantity / secondary_unit	secondaryQuantity / secondaryUnit	Optional pieces/meters/bags/bales; never auto-convert into kg.
listings.batch_id / owner_id / owner_organization_id / owner_name	batchId / ownerId / ownerOrganizationId / ownerName	Same batch, publishing owner and institution.
listings.category / condition / pathway / looking_for	textileType / quality / circularPathway / exchangeFor	looking_for array becomes semicolon-separated display string.
listings.weight_kg / available_weight_kg / reserved_kg	quantityKg; no independent reserved quantity in listing view	weight_kg takes precedence when present. Agree whether it means listed or currently available mass.
listings.weight_status / reward_points / is_demo / status	weightStatus / rewardCredits / isDemo / status	Integer points, explicit demo flag and canonical state translation.
listings.media	Not exposed by current BarterListing decoder	Stored mobile images are not automatically a web listing image gallery. Preserve JSON/media references during updates.

6.3 Offers transfers and evidence

Supabase column	Web alias	Meaning / matching rule
match_requests.id / listing_id / requester_id / provider_id	offer.id / listingId / recipientId / providerId	Requester is recipient; offer persistence uses match_requests.
requested_kg / incentive_points / status / terms	quantityKg / offeredCredits / status / terms	Proposal amount, proposed non-cash points and decision state.

Supabase column	Web alias	Meaning / matching rule
transfers.batch_id / listing_id / match_request_id	batchId / listingId / offerId	Connect all stages to original canonical records.
transfers.provider_id / recipient_id / agreed_weight_kg	providerId / recipientId / agreedWeightKg	Agreement participants and amount, not receipt weight.
transfers.circular_outcome / discrepancy_kg / discrepancy_reason	circularOutcome / discrepancyKg / discrepancyReason	Final pathway and documented signed variance.
transfers.pending_reward_points / reward_issued_at	pendingRewardCredits / rewardIssuedAt	Expected points and issuance marker; reconcile to reward_ledger.
measurements.transfer_id + stage handover / receipt / validation + weight_kg	handoverWeightKg / receiptWeightKg / verifiedWeightKg	Latest measured_at per stage supplies each transfer column.
measurements.batch_id, weight_status, method, recorded_by, verified_by, measured_at, note	batchId, status, method, recordedBy, verifiedBy, timestamp, note	Measurement evidence and accountable actors.
flow_events.transfer_id, stage, sequence, actor_id/name, occurred_at, event_status, measurement_id	transfer.events type/order/actorId/actorName/timestamp/status/measurementId	Structured timeline stored separately, enriched into transfer API response.

6.4 Recovery facilities rewards and engagement

Supabase column	Web alias	Meaning / matching rule
recovery_records.batch_id / transfer_id / facility_id / organization_id	batchId / transferId / facilityId / organizationId	Link source material, transfer and processing institution.
recovery_records.stage / input_kg / output_kg / rejected_kg	stage / inputWeightKg / outputWeightKg / residualWeightKg	Collection/Sorting/Processing/Recovery/Rejection balance. loss_kg exists but is not separately surfaced by current decoder.
recovery_records.status / actor_id / validated_by / created_at	status / recordedBy / validatedBy / timestamp	Recovery analytics use created_at, not a separate event-time field. Preserve meaning for backdated/offline entry.
facilities.categories / service_areas / monthly_capacity_kg / current_intake_kg	acceptedMaterials / serviceAreas / monthlyCapacityKg / currentIntakeKg	Registry configuration; analytics recompute intake from recovery records.
reward_ledger.user_id / transfer_id / points / entry_type / title	walletEntries.userId / transferId / credits / type / description	Ledger source; wallet_accounts balance is not current dashboard reward source.
system_interactions.event_name / occurred_at / client	eventName / occurredAt / client	Exact summary event vocabulary; client is web or mobile.
system_interactions.properties.count, averageSessionMinutes, region, city, synthetic	properties with the same keys	CamelCase averageSessionMinutes is intentional inside JSON. Counts and geography must be present.

Supabase column	Web alias	Meaning / matching rule
activity_logs.actor_id / action / entity_type / entity_id / details / occurred_at	actorId / action / entityType / entityId / details / timestamp	Audit evidence; not system_interactions or physical-flow counts.
configurations.key = platform / value	configuration.id = platform / typed rules	Database id commonly CFG-platform-v1. JSON value keys remain camelCase; do not query the business key as a literal row ID without the adapter.

7 Status and Lifecycle Contract

Entity	Persisted value	Web interpretation
Listing	available / matched / reserved / completed	Open / Matched / Reserved / Completed
Listing	draft / closed / archived	All display as Closed; finer mobile states are not distinguished.
Offer	pending / revised / accepted / rejected / withdrawn / expired	Pending / Pending / Accepted / Declined / Withdrawn / Declined
Transfer	reserved / in_progress	Reserved
Transfer	handed_over	Handed over; analytic transaction status In transit
Transfer	collected / received / processing	Received; analytic transaction status In sorting
Transfer	validated / completed	Validated / Completed; analytics maps Validated to Pending validation until completed
Transfer	cancelled / rejected / disputed	Cancelled / Rejected / Disputed; Rejected and Disputed analytic rows are Flagged
Batch	reused / upcycled / recycled / recovered	Recovered; circularPathway carries the detailed outcome
Measurement	declaration / listing; handover; receipt / collection; validation	Declaration; Handover; Receipt; Validation
Measurement	processing_input / recovery_output / rejection / loss	Recovery in measurement history; not separate analytic outcome totals
Recovery	recorded / validated / flagged / reversed	Recorded / Validated / Flagged / Reversed; reversed excluded from analytics
Account verification	verified; pending / rejected / suspended	Demo verified; all remaining values currently Pending demo review

Use the exact enum spelling and case from the PostgreSQL contract. Weight evidence is Estimated / Measured / Verified with capitals; recovery stage is Collection / Sorting / Processing / Recovery / Rejection; transfer status is lowercase. Unknown aliases are not normalized reliably. [S04-S05]

Normal governed transfer sequence: accepted proposal -> Reserved -> handover measurement/event -> Handed over -> receipt measurement/event and variance -> Received -> validation measurement/event -> Validated -> confirmed outcome and completion event -> Completed -> eligible reward ledger entry. Cancellation, rejection and dispute are explicit exceptions, not silent row deletion. [S01, S11]

A completed database status is not sufficient evidence by itself. Current API workflow creates the supporting records, but externally written rows can bypass that workflow. The analytic decoder chooses the latest validation-stage measurement without independently proving verifier authority, measurement status or supersession. Both developers must enforce those conditions in the authorized write path before treating results as verified.

8 Time Filters Units and Refresh

Concern	Current rule
Operational reporting clock	Asia/Manila calendar dates. Last 30/90 days include today through generation time; custom dates are inclusive; future/reversed/invalid ranges are rejected. Previous comparison is the immediately preceding equal elapsed interval, except YTD uses prior year to equivalent Manila date/time.
Cohort denominator	Batch created_at selects the material cohort. Linked latest outcomes through generation time are used even if completion occurs after the selected historical window. Do not compare a mobile completion-month chart with this registration-month chart without reconciling time basis.
Global material filters	Region/city, textile category, waste source, pathway, weight status, stakeholder role, producer/recycler search, organization, facility and transaction filters select related batches/transfers. Some filters require related user/recovery records to exist. Waste form is stored and displayed, but is not a global filter today.
Filter exceptions	Engagement uses reporting dates and region/city/access scope only. Listing oversight has its own local filters and all-time listing API load. Offer tab uses its own offer load. Predictive has dedicated fixed-snapshot filters. Administrative master lists are not automatically cohort-filtered.
Canonical mass	Persist and transmit kilograms in fields ending _kg or Kg. Display metric tons: 1 t = 1,000 kg. A 900 kg record displays as 0.9 t. Never save 0.9 into weight_kg when the intended material is 900 kg.
Other units and rounding	Pieces/meters/bags/bales remain secondary evidence with no automatic kg conversion. General percentages round to one decimal. Aggregate raw weights before rounding; exports retain up to six decimal places in tons.
Server caches	Dashboard source snapshot defaults to 15 seconds; calculated dashboard response defaults to 30 seconds, with actor/filter-aware keys. Successful recognized web API mutations invalidate both. External/mobile direct writes do not trigger this in-process invalidation.
Client cache	Dashboard cache is session-scoped with a 30-second freshness test when a load is triggered. Visited pages remain mounted. Same-tab mutations emit baliktela:records-changed and invalidate relevant data; sign-out clears private caches.
Important freshness limitation	A TTL is not a timer that refreshes the visible page. No operational real-time subscription or universal periodic data refresh is implemented. A mounted page can remain stale indefinitely until reload, applicable navigation/filter change, midnight key change, or local mutation triggers a fetch.
Authentication check	The 15-second foreground session check refreshes authorization visibility, not all dashboard datasets. Do not describe it as a 15-second mobile synchronization service.
Reliable comparison procedure	Wait for server cache expiration or use an authorized invalidation mechanism, reload/requery, and compare identical account scope, filters, batch cohort, unit and as-of timestamp. TTLs are not a guaranteed maximum synchronization latency.

9 Worked Reconciliation Example

Illustrative fixture only, not a production record. Select one source batch in one reporting cohort with declared mass 1,000 kg. It has Collection output 900 kg and Sorting output 850 kg at a facility with 2,000 kg prorated period capacity. Two non-overlapping completed outcomes are Reuse 300 kg and Recycling 450 kg, each supported by validation measurements.

Displayed information	Expected result	Why
Declared / generated	1 t	1,000 kg counted once, not once per transfer
Collected / sorted	0.9 t / 0.85 t	Different recovery stages, not additive material
Reused / recycled	0.3 t / 0.45 t	Completed outcome weights of distinct material portions
Recovered proxy	0.75 t	300 + 450 = 750 kg
Overall diversion / circularity	75%	750 / 1,000 x 100
Category recovery	83.3%	750 / 900 x 100, assuming one category
Sorting yield	88.2%	750 / 850 x 100
Residual / recovery gap	0.25 t	max(0, 1,000 - 300 - 450)
Facility utilization	45%	900 intake / 2,000 prorated capacity x 100
Completed transaction count	2	Two transfers, not five stage quantities or ten events

The total is not 1,000 + 900 + 850 + 300 + 450 kg. Those numbers describe overlapping stages and outcome portions of one original material cohort. Multiple transfers per batch require agreed allocation rules; the current normal API acceptance path reserves a listing and does not provide a complete arbitrary partial/split-batch workflow. The two-outcome illustration explains aggregation, not a promise that mobile can create any split without additional validation.

If previous recovered mass were 600 kg, the mass change would be +25%: (750 - 600)/600. If previous diversion rate were 70%, the rate movement would be +5 percentage points, not +5% relative growth. A missing denominator or missing measurement must not be disguised as a verified zero.

10 Tables That Do Not Automatically Feed Analytics

The canonical migration defines 36 public application tables. Table existence does not mean a current page reads or writes it. The current dashboard snapshot reads 11 domain datasets plus enriched flow_events; authentication/administration use additional tables. [S03, S05, S06, S13]

Table or group	Current web consumption / integration implication
collection_receipts	Not read by dashboard analytics. To affect collection totals, map receipt evidence into the agreed Collection recovery record and relevant measurements/events without counting the same mass twice.

Table or group	Current web consumption / integration implication
collection_events / event_registrations	Not joined into current regional activity, stakeholder participation or engagement totals. Event participation dashboard integration remains a separate task.
facility_assignments	Not used as the source of current users.scope access arrays. Assignment rows alone do not grant existing web permissions.
conversations / messages	Shared API operations exist; no current primary institutional dashboard messaging workspace is mounted. Message counts do not automatically become engagement totals.
favorites / notifications	Not current analytical inputs. Mobile actions here do not change reuse/recycling mass.
wallet_accounts / reward_rules	Not the current dashboard reward source/rate source. Web reads reward_ledger and platform configuration; coordinate rule versions and balances before treating mobile wallet totals as equivalent.
reports / support_tickets	reports means user-submitted moderation reports, not the analytical report catalog. These tables do not currently populate the web Reports page or a mounted support queue.
data_quality_issues	Current Data Quality findings are generated on demand; persistent issue acknowledgments/resolution are not sourced from this table.
methodologies / model_runs	Current governance uses configuration/fixed text; predictive models run from generated data in memory. These tables are not an active model registry or research approval workflow.
report_exports	Current CSV/print routes append activity_logs, not export-job records in this table. Do not look here for a complete current export history.
consent_history	Defined in contract; current surfaced consent values mainly come from users/privacy flags and profile timing. Full versioned consent-history consumption is not a current dashboard module.
Supabase Storage media	Referenced listing media can remain mobile-owned, but current oversight decoder does not render them. No storage bucket name or signed-URL contract can be inferred from this web decoder alone.

11 Mobile Developer Handoff and Acceptance

11.1 Agree on the write boundary

Preferred integration is one shared governed operational API so authorization, transitions, verification, rewards, audits and atomic writes have one owner. If mobile already uses separate RPCs/backend endpoints, retain that architecture only after both teams demonstrate the same canonical table contract and invariants. Do not call the service-role-only dashboard snapshot or management RPC directly from an untrusted client. Never place a Supabase secret/service-role key in either frontend.

Supabase identity and web application sessions are different token layers. The web API issues its own access JWT and database-backed application session after identity verification. A native Supabase access token is not automatically interchangeable with that JWT. Confirm the mobile authentication bridge or use the supported API login flow; a common Auth project alone does not prove API interoperability. [S01, S07, S26]

11.2 Minimum handoff packet

Mobile developer supplies	Web team checks
Supabase project/environment and actual table/RPC contract; no secrets in shared documents	Both apps target the same intended project/schema, not separate local/Atlas/production environments.
One sanitized user/profile and source organization/facility mapping	public.users.id vs auth.users.id; role/verification/account status; preserved organization and region access.
One linked batch -> listing -> proposal -> transfer record set	IDs match exactly; requester/provider direction correct; no duplicated source batch for relisted material.
Handover, receipt and validation measurements and flow events	Correct transfer_id/stage, measured_at/occurred_at, recorder/verifier, weights, ordering and discrepancy.
Collection / Sorting / Recovery stage balance records	Recovery stages and created_at semantics match the KPI reader; linked facility exists; reversed corrections do not duplicate outputs.
Reward event and idempotency policy	One issuance per eligible completion, points units/rate version, reversal/net balance meaning agreed.
Telemetry dictionary and sample summary rows	Exact event names, properties.count, averageSessionMinutes, disjoint MAU populations, region/city and privacy handling.
Geographic and taxonomy vocabulary	Use one standard string for each region/material. NCR and Metro Manila are currently different filter values; no alias conversion is implemented.
Client retry/offline conflict rules	Idempotent writes, preserved event time and canonical IDs, atomic stage updates, conflict responses and final-record immutability.

11.3 Controlled end to end test

Test	Acceptance evidence
Identity and visibility	Create a controlled synthetic participant in a staging/shared test scope. Confirm one profile bridge and correct institutional visibility; unrelated organization and aggregate-only viewers must not receive individual history.
Initial material entry	Create one known mobile batch/listing with a unique test namespace. Web listing oversight and material registry show the same batch ID, taxonomy and kilogram-equivalent amount.
Proposal and acceptance	Create a proposal, accept it once, retry safely. Exactly one reservation/transfer is produced, with provider/recipient and agreement preserved.
Physical lifecycle	Record handover, receipt and validation; confirm web stage columns equal mobile evidence and original declaration remains unchanged. Missing or unauthorized validation must not masquerade as verified.
Collection and sorting	Record explicit recovery balances; prove collection/sorting totals change once and regional/category totals reconcile to platform values. A receipt-only entry must be recognized as insufficient for current collection KPI.
Outcome and rewards	Complete with confirmed outcome; retry completion; verify no double-counted mass, no duplicate points and complete actor/timestamp event history.
Filters and reports	Use identical Manila date cohort, region/material and account scope. Match KPI, chart, regional/category sums and full CSV/print report at a controlled as-of timestamp.
Corrections and exceptions	Reverse/correct recovery with preserved original evidence; dispute/cancel unfinished transfer; ensure final records cannot be silently rewritten and quality findings remain explainable.
Freshness and access changes	Trigger actual re-fetch after server expiry and compare data; separately change role/scope to verify forced reauthentication. These are distinct acceptance tests.
Cleanup	Remove only disposable test records where policy permits, or preserve an agreed labeled fixture. Never run seeds, disable RLS, or modify unrelated records just to make tests pass.

Existing scripts/test-mobile-sync.ts is not sufficient proof of current mobile integration: it writes only a temporary batch, measurement and listing, uses a retired demo login, and uses legacy example taxonomy/location values. It does not validate offers, transfer completion, recovery, rewards, scopes or end-to-end analytics. Update that isolated test before running it against the shared project. No live synchronization test was executed for this report. [S21]

12 Reconciliation Work Still Required

The following are current interpretation or integration gaps, not changes made by this report. They should be agreed jointly before presenting mobile activity as complete operational evidence.

Priority	Gap and required agreement
High	Receipt/collection mismatch: collection_receipts and receipt measurements are not collected-mass inputs. Agree and implement the canonical conversion into deduplicated Collection recovery evidence.

Priority	Gap and required agreement
High	Outcome verification: completed status plus a validation-stage row is accepted by the reader; verifier authority, Verified evidence state, supersession and recovery validation must be enforced in the write boundary. Flagged/Recorded recovery rows currently still enter stage totals.
High	Partial material and repeated stages: analytics sum all qualifying recovery rows and completed outcomes; no universal repeated-stage/split/consolidation deduplication is inferred. Agree on unique physical allocations, correction reversals and child-batch lineage before multi-leg flows.
High	Telemetry: no automatic operational-action or session rollup to system_interactions exists. Define count aggregation and distinct-user de-duplication; raw mobile clicks without count will not populate existing engagement cards.
High	Freshness: no cross-client push invalidation/periodic operational refresh. Add an approved re-fetch trigger or governed real-time notification if near-live mobile visibility is required.
High	Authorization equivalence: a shared Auth account does not guarantee shared API token acceptance or safe mobile RPC permissions. Confirm the mobile write policy and current deployed RLS/RPC grants without exposing service secrets.
Medium	Geographic names and origin semantics: standardize NCR/Metro Manila and other aliases; agree that current mass is attributed to batch origin, not GPS or facility destination.
Medium	Demo provenance: dashboard builder hardcodes isDemo=true for transaction/stakeholder/facility analytical views and several UI labels always say sample/demo. Mixed synthetic and operational records are not globally segregated by an analytics demo filter. Preserve provenance and add appropriate segregation before real reporting.
Medium	Facility processing time is fixed 2.4 days; processed mass includes every completed outcome; status is stored. Replace/qualify these before claiming measured recycling throughput or lead times.
Medium	Outcome terminology differs: program diversion includes selected reuse/recycling pathways, while operational verified completion includes all completed outcomes. Diverted from Landfill is an operational proxy, not approved impact.
Medium	Active participants are region-level involvement counts, not unique platform users; verification aliases collapse rejected/suspended states; several empty-rate displays use zero. Current user roles classify historical participation, so a role change can also alter old cohort groupings. Clarify these rather than claiming historical role snapshots.
Medium	Barter Activity code/API fields exist but are not mounted in the current dashboard. Matches means proposal rows, and validated/completed/reward metrics are not limited to actual barter. Reward activity is earned points, not a net wallet balance.
Medium	Research consent and versioned approvals: current analytics do not select cohorts by research_consent and governance does not read approved methodology rows. Agree on consent/ethical-use requirements before real research extraction.
Medium	Measurement timing/supersession: latest measured_at wins; no automatic tie-breaker or supersedes_measurement_id exclusion in dashboard reads. Recovery timestamp comes from created_at. Coordinate offline/backdated records and corrections.
Planned	Collection events, persistent quality workflow, support/moderation records, model registry and export-job history have schema definitions but are not complete current dashboard feeds. Prioritize only the required web monitoring uses; mobile implementation remains with its developer.

13 Source Inventory and Reference Guide

References point to the reviewed repository commit so future changes do not silently change this report's evidence. The finalized blueprint in ../Documents/Balik Tela PH Web and Mobile Blueprint.docx establishes the management/analytics versus participation split, traceable stage weights, canonical kg, evidence-state distinction, no double counting and privacy requirements. Source code determines the implemented behavior described above; a blueprint requirement is not proof of implementation.

S01 | Runtime and API routes | server/index.ts:798

S02 | Dashboard cohort selection and calculations | server/dashboardAnalytics.ts:22

S03 | Supabase persistence and enrichment | server/supabaseStore.ts:12

S04 | Canonical field and status translation | server/database/codecs.ts:19

S05 | Actual PostgreSQL table definitions | supabase/migrations/20261007040000_baliktela_schema_v2.sql:17

S06 | Dashboard source snapshot SQL | supabase/migrations/20261012120000_dashboard_snapshot_rpc.sql:6

S07 | Role permissions and record scopes | server/access-control.ts:28

S08 | Reporting dates and capacity proration | shared/reporting.ts:50

S09 | Units and percentage helpers | shared/metrics.ts:10

S10 | Data quality rules | server/dataQuality.ts:4

S11 | Atomic management workflows | server/management.ts:65

S12 | Reports and exports | server/reports.ts:29

S13 | Web routes and mounted modules | src/DashboardApp.tsx:68

S14 | Web data cache and invalidation | src/hooks/useDashboardData.ts:11

S15 | Synthetic operational source builder | server/analytics/operationalDemo.ts:8

S16 | October synthetic scenario | server/analytics/octoberDemo.ts:7

S17 | Predictive synthetic generator | server/analytics/synthetic.ts:18

S18 | Predictive methods and evaluation | server/analytics/predictive.ts:134

S19 | Predictive endpoints and export scope | server/analytics/routes.ts:25

S20 | Shared authentication identity bridge | server/provisioning.ts:4

S21 | Legacy synchronization test limits | scripts/test-mobile-sync.ts:25

S22 | Geographic reference assets | src/lib/geography.ts:1

S23 | Overview and rule-based insight text | src/pages/Overview.tsx:15

S24 | Authenticated operational mass accounting | server/domain.ts:7

S25 | Original synthetic figures | src/data/mockData.ts:1

S26 | Supabase identity provider | server/supabaseIdentity.ts:16

S27 | Predictive dataset dictionary | data/predictive/v1/data-dictionary.json:1

S28 | Public homepage content | src/data/publicContent.ts:1

Appendix A Canonical Table Coverage

Table	Current relationship to the web dashboard
schema_versions	Schema contract metadata; not a KPI input.
users	Identity, profile, approval, role/scope; joins for participation and attribution.
auth_account_security	Email/password lifecycle; session security; not user engagement.
organizations	Institution master data, lookup/filtering and participation reports.
sessions	Revocable application authentication sessions; not current engagement summary source.
auth_challenges	Single-use account setup/recovery authorization; not analytics.
auth_rate_limits	Authentication abuse prevention; not public usage statistics.
consent_history	Defined; no complete current dashboard history reader.
facilities	Facility master data, capacity and geography; operational/analytical views.
facility_assignments	Defined; current web authorization uses users.scope arrays.
collection_events	Defined; not joined into current analytics.
event_registrations	Defined; not joined into current analytics.
material_batches	Primary declared material cohort and geographic/category denominator.
measurements	Stage evidence and latest handover/receipt/validation weights.
listings	Oversight and listing counts; linked to batches.
favorites	Defined mobile participation data; not current dashboard input.
match_requests	Web offers; proposal records and transfer agreement link.
transfers	Agreement, status, outcome and verified completed mass attribution.
flow_events	Transfer lifecycle timeline enrichment.
collection_receipts	Defined receipt source; not directly included in collection KPI.
recovery_records	Collection/sorting balances, facility attribution and recovery monitoring.
conversations	Shared API exists; no mounted institutional messaging page.
messages	Shared API exists; not an engagement rollup input.

Table	Current relationship to the web dashboard
notifications	Defined; not current analytical input.
reward_rules	Defined; current completion rate is platform configuration.
reward_ledger	Reward/wallet history and earned credits in analytics.
wallet_accounts	Defined balance snapshot; not current analytic reward source.
reports	Moderation submissions, not analytical report definitions.
support_tickets	Defined; no mounted support-case dashboard reader.
activity_logs	Administrative/authentication/transaction/export audit trail.
configurations	Active platform rules, taxonomies and methodology version.
data_quality_issues	Defined; current quality findings are computed on request.
system_interactions	Count-summary source for engagement analytics.
methodologies	Defined; not current governance approval source.
model_runs	Defined; not current predictive runtime persistence.
report_exports	Defined; current report exports append activity_logs instead.

Appendix B Database Column Names

The following inventory is extracted from the checked-in initial PostgreSQL migration, not inferred from UI labels. Later migrations add authentication functions, session-revocation behavior and transaction RPCs. This inventory is a baseline handoff, not a live information_schema export. The legacy mixed-case passwordHash column is credentials compatibility metadata; use Supabase Auth, never a client-side password table write.

schema_versions

id, version, status, applied_at, applied_by, notes

users

id, auth_user_id, email, passwordHash, name, username, role, active, stakeholder, organization_id, organization, organization_verified, verification_status, account_status, bio, location, city, region, phone, avatar_uri, rating, swaps, privacy_consent, research_consent, notification_enabled, scope, profile, trust_placeholder, last_login_at, created_at, updated_at

auth_account_security

id, user_id, email_verified_at, password_changed_at, created_at, updated_at

organizations

id, name, type, verification_status, status, record_label, region, city, address, service_areas, accepted_materials, contact_email, created_at, updated_at

sessions

id, user_id, client, token_hash, used_token_hashes, csrf_hash, expires_at, revoked_at, created_at, updated_at

auth_challenges

id, user_id, type, token_hash, client, expires_at, consumed_at, created_at, updated_at

auth_rate_limits

id, scope, subject_hash, attempts, window_started_at, locked_until, expires_at, updated_at

consent_history

id, user_id, privacy, research, policy_version, recorded_by, source, created_at

facilities

id, organization_id, name, type, address, city, region, location, categories, pathways, schedule, instructions, service_areas, monthly_capacity_kg, current_intake_kg, is_demo, verification_status, status, created_at, updated_at

facility_assignments

id, user_id, facility_id, role, active, assigned_by, created_at, updated_at

collection_events

id, facility_id, title, starts_at, ends_at, description, status, created_by, approved_by, approved_at, created_at, updated_at

event_registrations

id, event_id, user_id, status, created_at, updated_at

material_batches

id, source_user_id, source_organization_id, material_category, composition, waste_source, waste_form, condition, intended_pathway, final_pathway, declared_weight_kg, current_weight_kg, weight_status, secondary_quantity, secondary_unit, location, current_status, current_custodian_id, validation_status, is_demo, archived_at, created_at, updated_at

measurements

id, batch_id, transfer_id, facility_id, flow_event_id, stage, weight_kg, weight_status, method, measurement_context, recorded_by, verified_by, measured_at, note, supersedes_measurement_id, created_at

listings

id, batch_id, owner_id, owner_organization_id, owner_name, title, description, looking_for, media, available_weight_kg, reserved_kg, category, condition, location, city, region, weight_kg, weight_status, waste_source, waste_form, pathway, reward_points, is_demo, status, published_at, created_at, updated_at

favorites

id, user_id, listing_id, created_at

match_requests

id, requester_id, requester_organization_id, listing_id, listing_title, requester_name, provider_id, requested_kg, terms, incentive_points, status, version, previous_terms, responded_by, responded_at, created_at, updated_at

transfers

id, listing_id, batch_id, match_request_id, provider_id, provider_organization_id, recipient_id, recipient_organization_id, material_category, agreed_weight_kg, transfer_type, status, agreed_terms, circular_outcome, discrepancy_kg, discrepancy_reason, pending_reward_points, reward_issued_at, is_demo, validation_status, completed_at, cancelled_at, created_at, updated_at

flow_events

id, batch_id, listing_id, transfer_id, actor_id, actor_name, facility_id, stage, sequence, event_status, weight_kg, weight_status, measurement_id, location, note, occurred_at, created_at

collection_receipts

id, transfer_id, batch_id, facility_id, facility_name, source_user_id, actor_id, measurement_id, weight_kg, weight_status, measurement_context, note, created_at

recovery_records

id, transfer_id, batch_id, facility_id, organization_id, actor_id, validated_by, stage, input_kg, output_kg, rejected_kg, loss_kg, outcome, status, is_demo, note, created_at, updated_at

conversations

id, listing_id, transfer_id, participant_ids, participant_names, created_at, updated_at

messages

id, conversation_id, sender_id, sender_name, body, read_by, created_at, edited_at

notifications

id, user_id, type, title, body, entity_type, entity_id, is_read, read_at, created_at

reward_rules

id, title, points, active, version, eligibility, created_by, approved_by, created_at, updated_at

reward_ledger

id, user_id, transfer_id, rule_id, points, entry_type, title, idempotency_key, created_at

wallet_accounts

id, user_id, balance_points, updated_at

reports

id, reporter_id, target_type, target_id, reason, details, status, assigned_to, resolution, created_at, updated_at

support_tickets

id, user_id, category, subject, details, attachment_uri, status, assigned_to, created_at, updated_at

activity_logs

id, actor_id, action, entity_type, entity_id, details, ip_hash, occurred_at

configurations

id, key, version, status, value, effective_from, updated_by, created_at, updated_at

data_quality_issues

id, entity_type, entity_id, rule_code, severity, status, description, evidence, detected_at, assigned_to, resolved_by, resolved_at, resolution_note, updated_at

system_interactions

id, user_id, organization_id, session_id, event_name, client, properties, privacy_classification, occurred_at

methodologies

id, name, type, version, status, definition, limitations, approved_by, approved_at, created_at, updated_at

model_runs

id, methodology_id, status, target, input_period, filters, output, metrics, created_by, created_at, completed_at

report_exports

id, requested_by, report_type, scope, privacy_classification, status, row_count, file_uri, requested_at, generated_at, expires_at

Integration conclusion: match canonical operational records and evidence first, then compare derived totals using identical definitions. Do not match an unrelated mobile card to a similarly named web card until record grain, stage, unit, period, outcome and access scope are the same.